TOPIC 11

Common Tests and Choosing One

Two-sample and paired t-tests, chi-square, ANOVA, non-parametric alternatives, effect sizes and a decision guide

WHAT YOU WILL LEARN

- Choose the right test from the type of data and the question
- Run and interpret two-sample, paired, chi-square and one-way ANOVA tests
- Know when to switch to non-parametric tests
- Check assumptions and report effect sizes
- Write up a result correctly

1. A map of the common tests

Every test in this topic follows the six steps of Topic 10: hypotheses, α, assumptions, test statistic, p-value, decision. Only the **statistic** and its **reference distribution** change. Choose by asking: *what type of data do I have, and how many groups am I comparing?*

Question	Data	Test	Statistic / distribution
Is one mean equal to a value?	Numerical	One-sample t-test	t, df = n − 1
Do two **independent** groups differ in mean?	Numerical	Two-sample (Welch) t-test	t, Welch df
Did the **same** subjects change?	Numerical, paired	Paired t-test	t on differences
Do 3 or more groups differ in mean?	Numerical, one factor	One-way ANOVA	F
Do observed counts fit a model?	Categorical counts	Chi-square goodness of fit	χ^2, df = k − 1
Are two categorical variables related?	Contingency table	Chi-square test of independence	χ^2, df = (r − 1)(c − 1)
Compare means but data are non-normal or ordinal	Numerical / ordinal	Mann-Whitney, Wilcoxon, Kruskal-Wallis	Rank-based
Is a proportion equal to a value?	Binary	One-proportion z-test	z

2. Two independent samples: the Welch t-test

Compare the means of two **separate** groups (for example, two teaching methods). H_0: $\mu_1 = \mu_2$ (equivalently $\mu_1 - \mu_2 = 0$).

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Welch's version does not assume equal variances and is the safe default. This is the same standard error you met in Topic 07, and the same t-statistic whose confidence interval appeared

in Topic 09.

> **WORKED EXAMPLE** **Two teaching methods (from Topic 09)**
> Method A: n = 40, $\bar{x}$ = 82.4, s = 9.1. Method B: n = 35, $\bar{x}$ = 78.1, s = 10.3. Two-sided test at α = 0.05. t = **1.904**, p = **0.0611**. Since p > 0.05 we **fail to reject H₀**. This matches the Topic 09 interval [−0.21, 8.81], which just includes 0. The effect size is Cohen's d = 0.44 (small to medium); the study may simply be too small to detect it.

3. Paired samples: the paired t-test

When the two measurements are on the **same units** (before and after, left and right) they are not independent. Compute the **difference** for each pair and run a one-sample t-test on the differences with H₀: μ_d = 0. Pairing removes person-to-person variation, so it is usually far more powerful.

> **WORKED EXAMPLE** **Blood pressure before and after treatment**
> Eight patients' systolic pressure: before = [140, 152, 138, 147, 160, 155, 149, 143], after = [132, 147, 136, 140, 151, 150, 145, 139]. Differences (before − after) = [8, 5, 2, 7, 9, 5, 4, 4]; mean difference $\bar{d}$ = 5.50, s_d = 2.33. Paired t = $\bar{d}/(s_d/\sqrt{8})$ = **6.677**, df = 7, p = **0.0003**, so we **reject H₀**. A 95% CI for the mean drop is [3.55, 7.45].

For comparison, ignoring the pairing with a two-sample test gives p = 0.149: the effect disappears into between-patient variation. Always use the paired test for paired data.

4. Chi-square tests for counts

Chi-square tests compare **observed counts** (O) with **expected counts** (E) under H₀.

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Large χ² means the data are far from what H₀ predicts. Requirement: every expected count should be at least about 5.

Goodness of fit: is the die fair?

Face	1	2	3	4	5	6
Observed	25	17	15	23	24	16
Expected (H₀: fair)	20	20	20	20	20	20
(O − E)²/E	1.25	0.45	1.25	0.45	0.80	0.80

Total χ² = **5.00** with df = 6 − 1 = 5, p = **0.416**. We **fail to reject H₀**: 120 rolls with this much variation are quite consistent with a fair die.

Independence: are two categorical variables related?

Test whether the outcome depends on the group. Expected count for each cell = (row total × column total)/grand total, and df = (rows − 1)(columns − 1).

	Success	Failure	Total
New treatment	60	40	100
Standard treatment	45	55	100
Total	105	95	200

Expected counts if treatment and success were independent: 52.5, 47.5 / 52.5, 47.5. χ^2 = **4.511**, df = 1, p = **0.0337**, so we **reject H_0**: the success rate differs between treatments (60% versus 45%). (Software applies a continuity correction to 2×2 tables by default; here it moves p from 0.034 to 0.047, still below 0.05 but closer, so say which version you used.)

> **COMMON MISTAKE** **Association is not causation**
> A significant chi-square says the variables are **related**. Only a well-designed randomized experiment can support a causal claim.

5. One-way ANOVA: comparing three or more means

Running many pairwise t-tests inflates the false-positive rate (Topic 10). **Analysis of variance (ANOVA)** tests all the group means at once. H_0: $\mu_1 = \mu_2 = \ldots = \mu_k$; H_1: at least one mean differs.

The idea: compare the variation **between** group means with the variation **within** groups.

$$F = \frac{\text{variance between groups (MSB)}}{\text{variance within groups (MSW)}} = \frac{SSB/(k-1)}{SSW/(N-k)}$$

Group	Scores	Mean
A	78, 82, 85, 80, 79, 84	81.33
B	72, 75, 70, 74, 77, 73	73.50
C	80, 83, 79, 85, 81, 78	81.00

Source	SS	df	MS	F	p
Between groups	235.44	2	117.72	17.17	0.00013
Within groups	102.83	15	6.86		
Total	338.28	17			

F(2, 15) = **17.17**, p = **0.00013**, so we **reject H_0**: the means are not all equal. ANOVA does not say *which* differ; a **post-hoc test** does (Tukey HSD controls the overall error rate): A vs B p = 0.0003, A vs C p = 0.9736, B vs C p = 0.0005. Group B is lower than both others, while A and C are not clearly different.

Assumptions: independent observations, roughly normal groups, similar variances. ANOVA is fairly robust with equal group sizes.

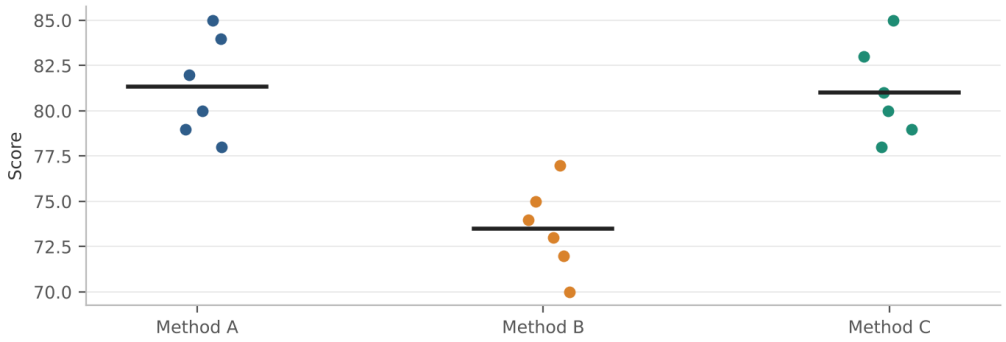


Figure 1. Scores by group; black bars are group means. The spread within each group is small compared with the gap to group B, giving a large F.

6. When the assumptions fail: non-parametric tests

The t-tests and ANOVA assume roughly normal data (or large n). With small samples of skewed or ordinal data use **rank-based** alternatives. They compare ranks rather than raw values, so outliers matter much less, at a small cost in power when the data really are normal.

Parametric test	Non-parametric alternative	Compares
One-sample / paired t-test	Wilcoxon signed-rank test	Median of differences
Two-sample t-test	Mann-Whitney U (Wilcoxon rank-sum)	Whether one group tends to be larger
One-way ANOVA	Kruskal-Wallis test	Groups' rank distributions
Pearson correlation	Spearman rank correlation	Monotonic association

Example: two small skewed samples (the first has an outlier at 12.0). Welch's t-test gives $p = 0.228$, while Mann-Whitney U gives $p = 0.026$. The outlier inflates the variance in the first group and weakens the t-test; ranks are not fooled by it.

7. Assumptions and effect sizes

Test	Key assumptions	How to check
t-tests	Independence; roughly normal data or n large; (Welch: no equal-variance need)	Histogram, Q-Q plot, outlier check
ANOVA	Independence; normality in each group; similar variances	Residual plots, Levene's test
Chi-square	Independent observations; expected counts ≥ 5	Look at the expected table

A p-value says whether an effect is detectable; an **effect size** says how big it is.

Effect size	For	Rough guide
Cohen's d = $(\bar{x}_1 - \bar{x}_2)/s_{pooled}$	Two means	0.2 small, 0.5 medium, 0.8 large
$\eta^2 = SSB/SST$	ANOVA	0.01 small, 0.06 medium, 0.14 large
Cramér's V	Chi-square tables	Depends on table size
Risk difference / odds ratio	Binary outcomes	Context-specific

For the ANOVA above, $\eta^2 = 0.70$: about 70% of the variation in scores is explained by method, a large effect.

8. Writing up a result

KEY IDEA **A good report contains**

- The test used and why.
- The statistic with its degrees of freedom and the p-value.
- An effect size and a confidence interval.
- A conclusion in plain language about the original question.

WORKED EXAMPLE **Example**

'A paired t-test showed that systolic blood pressure fell significantly after treatment (mean drop 5.5 mmHg, 95% CI [3.6, 7.4], $t(7) = 6.68$, $p < 0.001$, $n = 8$).'

9. A decision guide

Start here	Then	Use
Outcome is numerical	1 group vs a value	One-sample t-test
	2 independent groups	Welch t-test (Mann-Whitney if non-normal)
	2 paired measurements	Paired t-test (Wilcoxon if non-normal)
	3+ groups	One-way ANOVA (Kruskal-Wallis if non-normal), then post-hoc
Outcome is categorical	1 variable vs a model	Chi-square goodness of fit (or z-test for one proportion)
	2 variables, table of counts	Chi-square test of independence (Fisher's exact if counts are small)

10. In Python

```
from scipy import stats
import numpy as np

# Two independent groups
stats.ttest_ind(a, b, equal_var=False)          # Welch t-test
stats.mannwhitneyu(a, b)                        # non-parametric version

# Paired
stats.ttest_rel(before, after)
stats.wilcoxon(before, after)

# Categorical
stats.chisquare(observed, expected)             # goodness of fit
stats.chi2_contingency(table)                   # independence test

# 3+ groups
stats.f_oneway(g1, g2, g3)
stats.kruskal(g1, g2, g3)
stats.tukey_hsd(g1, g2, g3)                     # post-hoc comparisons
```

11. Where to go next

This handbook ends where modelling begins. Natural next topics are **correlation and linear regression** (testing relationships between numerical variables, using the same t and F ideas), **logistic regression** for binary outcomes, **power analysis and experimental design** (A/B testing), **non-parametric and resampling methods** (bootstrap, permutation tests) and **Bayesian inference**, which treats parameters as random and answers 'what is the probability that H_0 is true?' directly.

SUMMARY

- Choose the test from the data type and the design (one group, two independent, paired, three or more, counts).
- Welch t-test for two independent means; paired t-test for before/after; ANOVA for three or more means.
- Chi-square compares observed with expected counts: goodness of fit and independence.
- Rank-based tests are the safe choice for small, skewed or ordinal data.
- Always report effect size and confidence interval, not just the p-value.

Practice questions

Try each question before reading the answer underneath it.

Q1. Fifty customers rate satisfaction before and after a redesign. Which test?

Answer. Paired t-test (or Wilcoxon signed-rank if the differences are far from normal), because the same customers are measured twice.

Q2. Mean delivery time is compared across 4 couriers. Which test and why not 6 t-tests?

Answer. One-way ANOVA. Six pairwise tests inflate the overall false-positive rate (about $1 - 0.95^6 = 26\%$).

Q3. A 3 × 2 table has expected counts all above 5. Give df for a chi-square test of independence.

Answer. $(3 - 1)(2 - 1) = 2$.

Q4. A two-sample t-test with n = 3,200 per group gives p = 0.045, but Cohen's d = 0.05. Comment.

Answer. Statistically significant ($p < 0.05$) but the effect is tiny ($d = 0.05$, far below even a 'small' 0.2), so it is probably not practically important. The large n made it detectable.

Q5. ANOVA gives p = 0.001. Can you say all means differ?

Answer. No, only that at least one differs. Follow with a post-hoc test such as Tukey HSD.

Q6. Two small samples are heavily skewed with outliers. Which test would you use for a difference?

Answer. Mann-Whitney U, since it is rank-based and robust to outliers.